

automatically started when another NFC device enters the previously-specified range. Once in range, the two devices instantly communicate and send prompts to the user to initiate mobile payments, data sharing or wireless charging.

NFC wireless charging mode allows contactless transfer of up to 1W power to small devices like Bluetooth headset, fitness tracker or smartwatch, and other IoT devices.

Zigbee 3.0

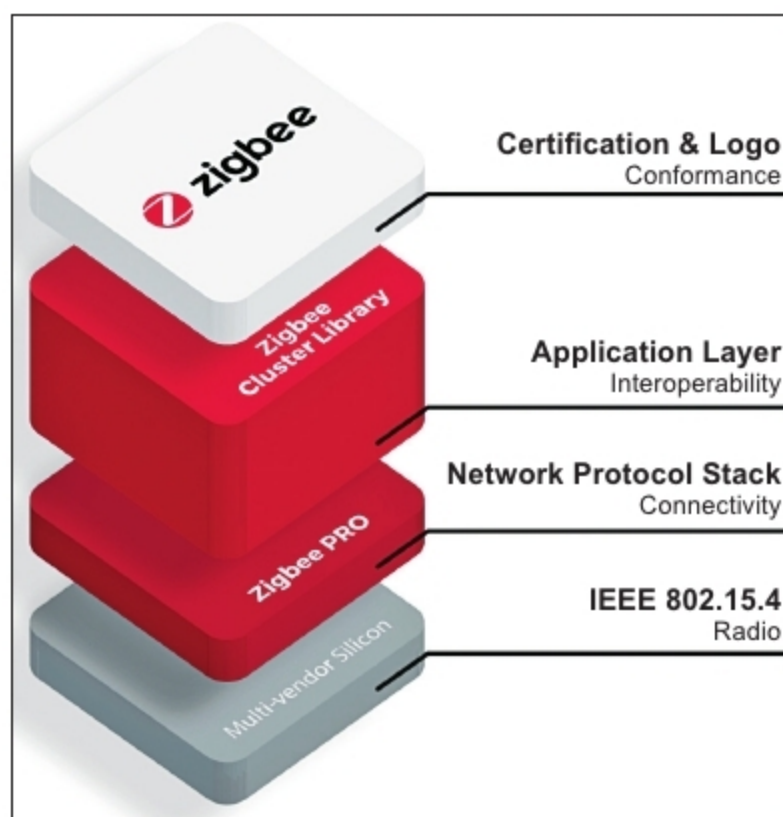
Zigbee is a standard wireless communication protocol that defines physical and Media Access Control (MAC) layers to handle many devices at low data rates. It uses 802.15.4 transceiver as base and operates at 868MHz, 902MHz to 928MHz, and 2.4GHz frequencies. It can build large mesh networks for device monitoring and control, handling up to 65,000 nodes. It can also support multiple types of radio networks such as point-to-point and point-to-multipoint. It can transfer data wirelessly over a distance of up to hundred metres at 250kB/s speed.

Zigbee technology is best suited for low-latency and low-energy consumption at lower bandwidths. It can transfer data in two modes: beacon and non-beacon. When there is no data communication from end devices, coordinators and routers in a beacon mode enter into sleep state.

Here, coordinators are responsible for initiating and managing devices over the network. They wake up after regular intervals and transmit beacons to the routers in the network. Thus, they operate when communication is needed, resulting in lower duty cycles and longer battery usage.

In non-beacon mode, coordinators and routers continuously monitor the active state of incoming data and, hence, consume more power.

Zigbee can be used for a range of applications including remote patient monitoring, wireless lighting and



Zigbee 3.0 (Credit: <https://zigbee.org>)

electrical meters, traffic management systems, industrial automation, home automation and smart grid monitoring. Smart grid monitoring involves remote temperature monitoring, fault locating, reactive power management and so on.

It also helps retailers operate more efficiently by supporting just-in-time inventory practices, as well as monitoring temperatures, humidity, spills and so on.

The latest Zigbee 3.0, built on the existing Zigbee standard, unifies the market-specific application to allow all devices to be wirelessly connected in the same network.

A Zigbee 3.0 certification scheme ensures interoperability of products from different manufacturers and 128-bit AES encryption for secure data connections.

RF transmission and SDR

RF is a form of electromagnetic transmission, ranging from 3kHz to 300GHz. It is used in wireless communication due to its property to penetrate through objects and travel long distances. Communication range depends on wavelength, transmitter power, receiver quality, and type, size and height of the antenna.

RF uses one transmitter section followed by a transmitting antenna, which is responsible for wireless signal transmission. This transmitted signal is received by the receiving antenna, amplified and demodulated to the original signal. This technology is widely used in military communications, personal communications, radio broadcast and communications in workplaces.

Software-defined radio (SDR) is a radio communication system

that utilises software to implement components that have been added in the hardware transmitter/receiver unit on a computer or embedded system. SDR uses software to perform many of the basic functions on signals like mixing, amplification, filtering, modulation/demodulation, detection, etc.

The software is easy to reconfigure and can be used on many platforms. Using the power of digital processing, SDR is being used for different applications in many different areas. The software is used to change the configuration of the radio or upgrade as per the function required at a given time.

It is possible to apply changes to any standard and even add new waveforms purely by upgrading the software and without the need for changes to the hardware. This can even be done remotely, thereby providing considerable savings in cost. It is also possible to reconfigure radios and change their performance by updating software.

Satellite communication

Microwaves are a form of electromagnetic transmission used in wireless communication systems. Wavelength of microwaves ranges